



# Muscle tissue and vessels

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# MUSCLE TISSUE

A unique feature of muscle cells (or fibers) with high content of proteins called *myofibrils*, which contract after stimulation of the muscle fibers. The myofibrils are composed of short protein threads, the *myofilaments*. These can be divided into thin *actin filaments* and thick *myosin filaments*.

Muscle tissue is divided into *smooth* and *striated muscle tissue* according to structure and function.

# General features

- Less extracellular matrix
- High energy demand – *lot of mitochondria*
- Myofilaments
- High calcium demand – *sarcoplasmic reticulum,  $Ca^{2+}$ -pumps and channels*

# Filaments

## Thin filaments:

diameter: approx. 6-8 nm , length: approx. 1  $\mu\text{m}$

ACTIN PROTEINS: F, G

## Thick filaments:

diameter: approx. 15 nm, length: kb. 1,5  $\mu\text{m}$

MYOSIN II. PROTEIN

*all thick filaments are built up by 200-300 myosin II. molecules*

The cytoplasm of muscle cells (=sarcoplasm) represents mainly actin and myosin II. proteins.

Actin and myosin proteins are also presented in the most cell types (e.g. migrating cells, phagocytic cells and glandular cells).

# CLASSIFICATION OF MUSCLE TISSUES

## STRIATED MUSCLE:

- SKELETAL MUSCLE
- VISCERAL STRIATED MUSCLE

*(tongue, pharynx, upper part of the esophagus, lumbar part of the diaphragm) – speaking, swallowing and respiration*

- CARDIAC MUSCLE

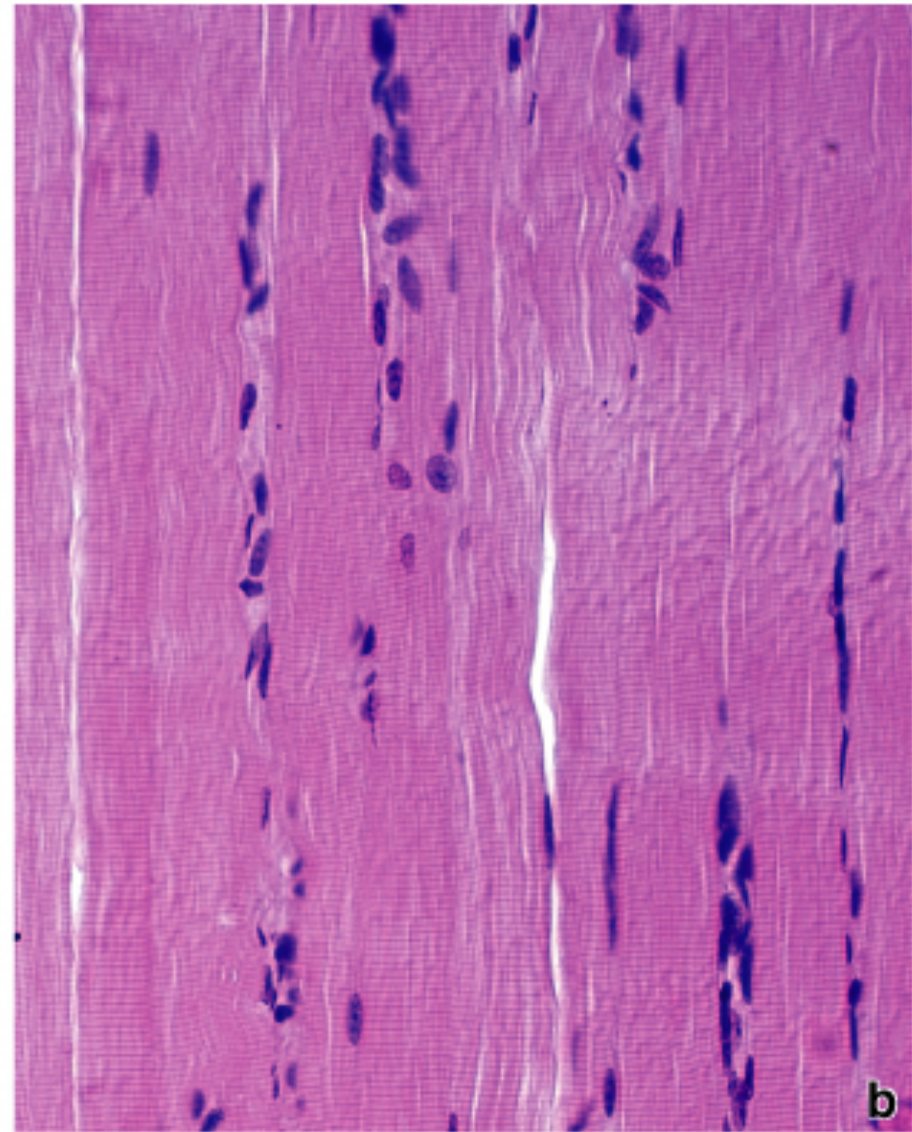
## SMOOTH MUSCLE:

*(wall of vessels and organs, internal eye muscles, arrector pili muscle)*

# SKELETAL MUSCLE

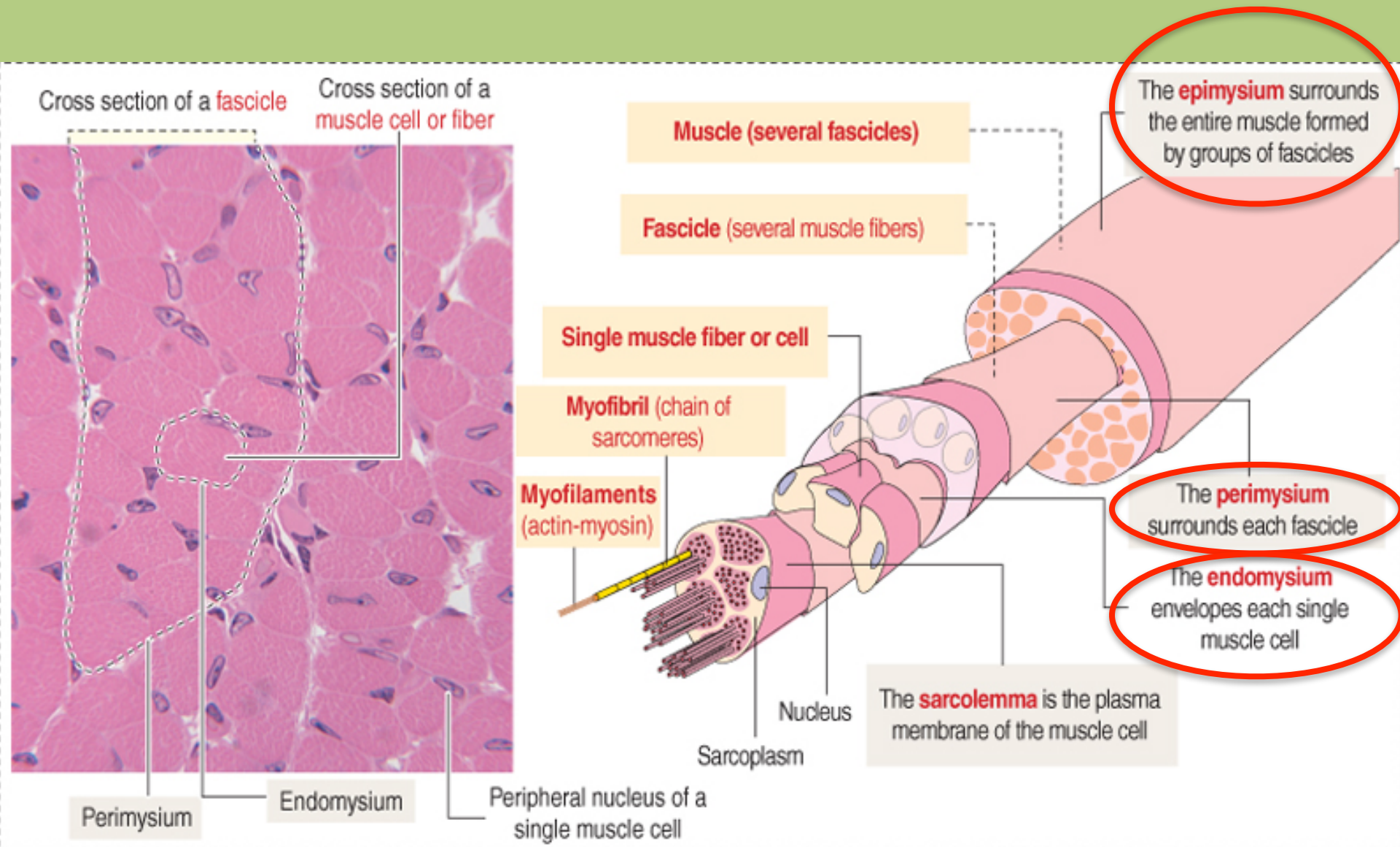
- Multi-nucleated cells
- The nuclei are located right under the cell membrane (*sarcolemma*)
- Length: from a few millimetres to one meter

# SKELETAL MUSCLE



**Figure 10.1b. Higher magnification photomicrograph of a skeletal muscle. X360.**

# Connective tissue coverings



# STRIATION

**A band:** myosin

**I band:** actin

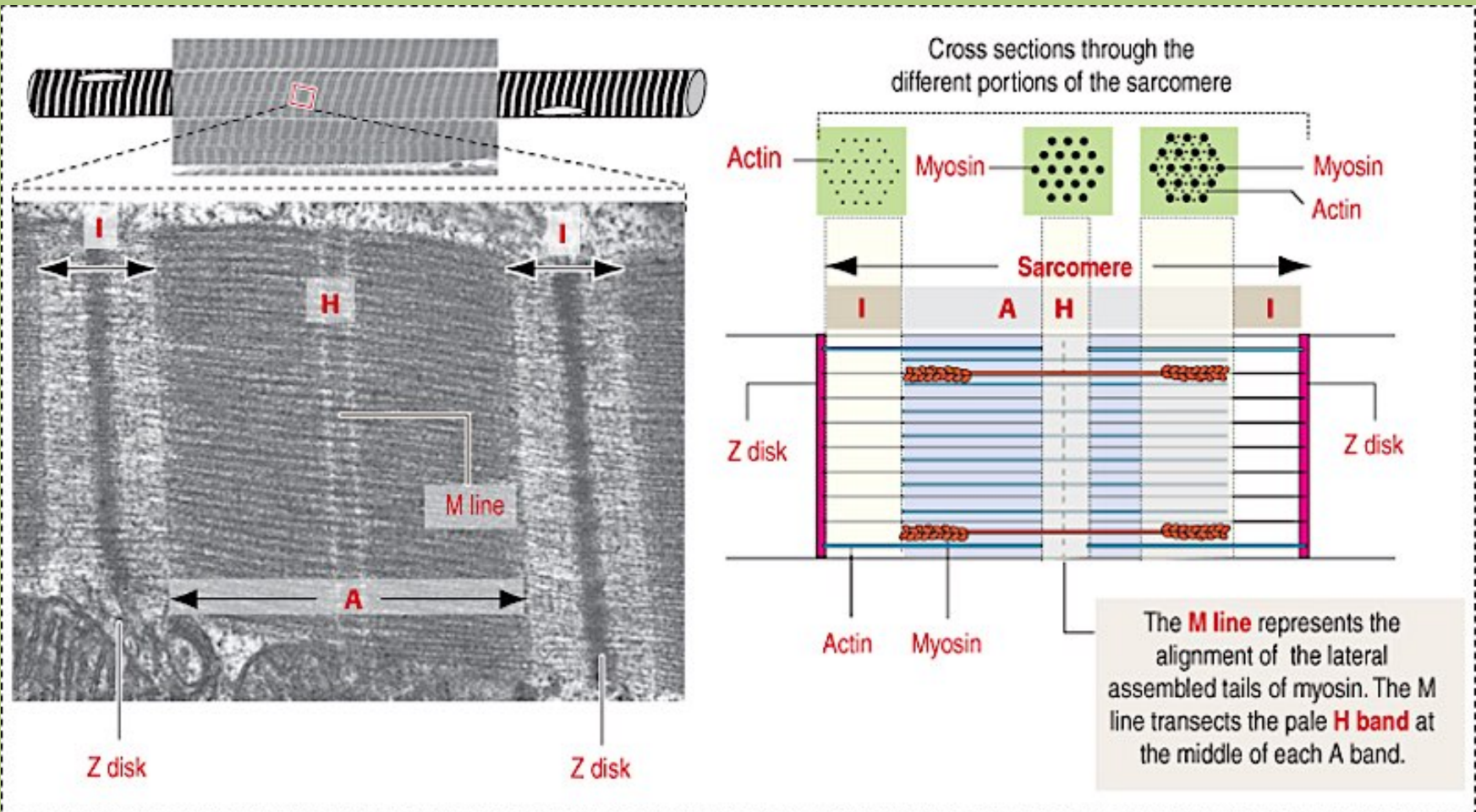
**Z line:** actin filaments

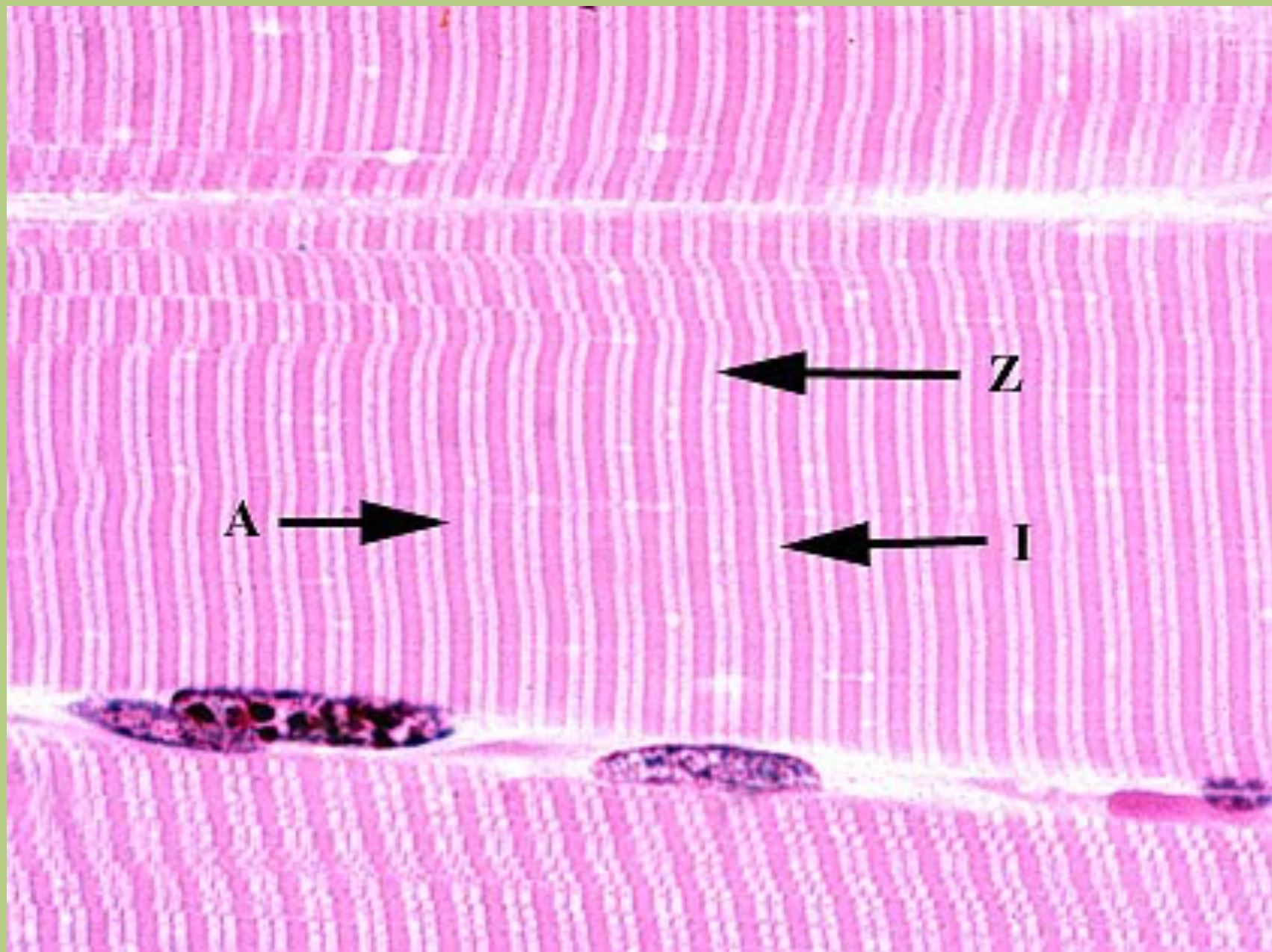
connect to it

**H band:** only myosin

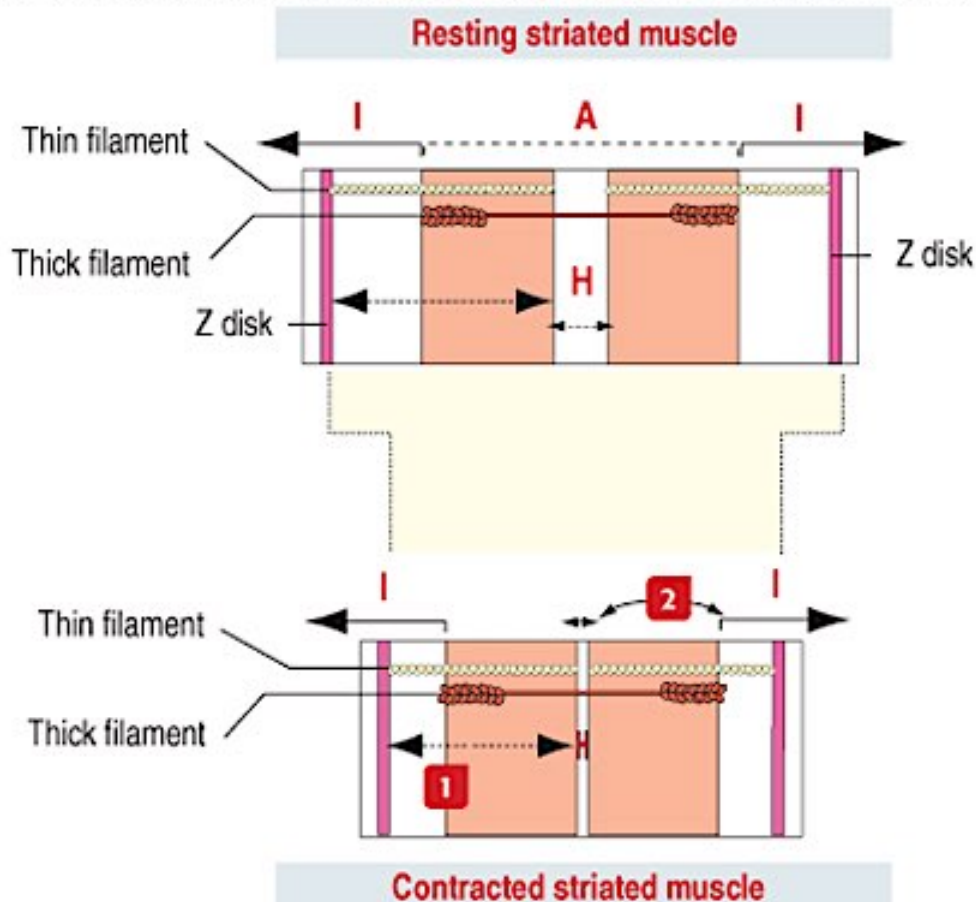
**M line:** myosin binding C proteins

**Sarcomere:** unit of the muscle tissue;  
distance between two Z lines





# SLIDING FILAMENT THEORY



The A band represents the distribution of the myosin thick filaments. The H band represents the myosin tail regions of the thick filaments not overlapping with thin filaments.

Thin filaments are attached to the Z disk. Two half-I bands, containing thin filaments, are seen at the right and left side of the Z disk.

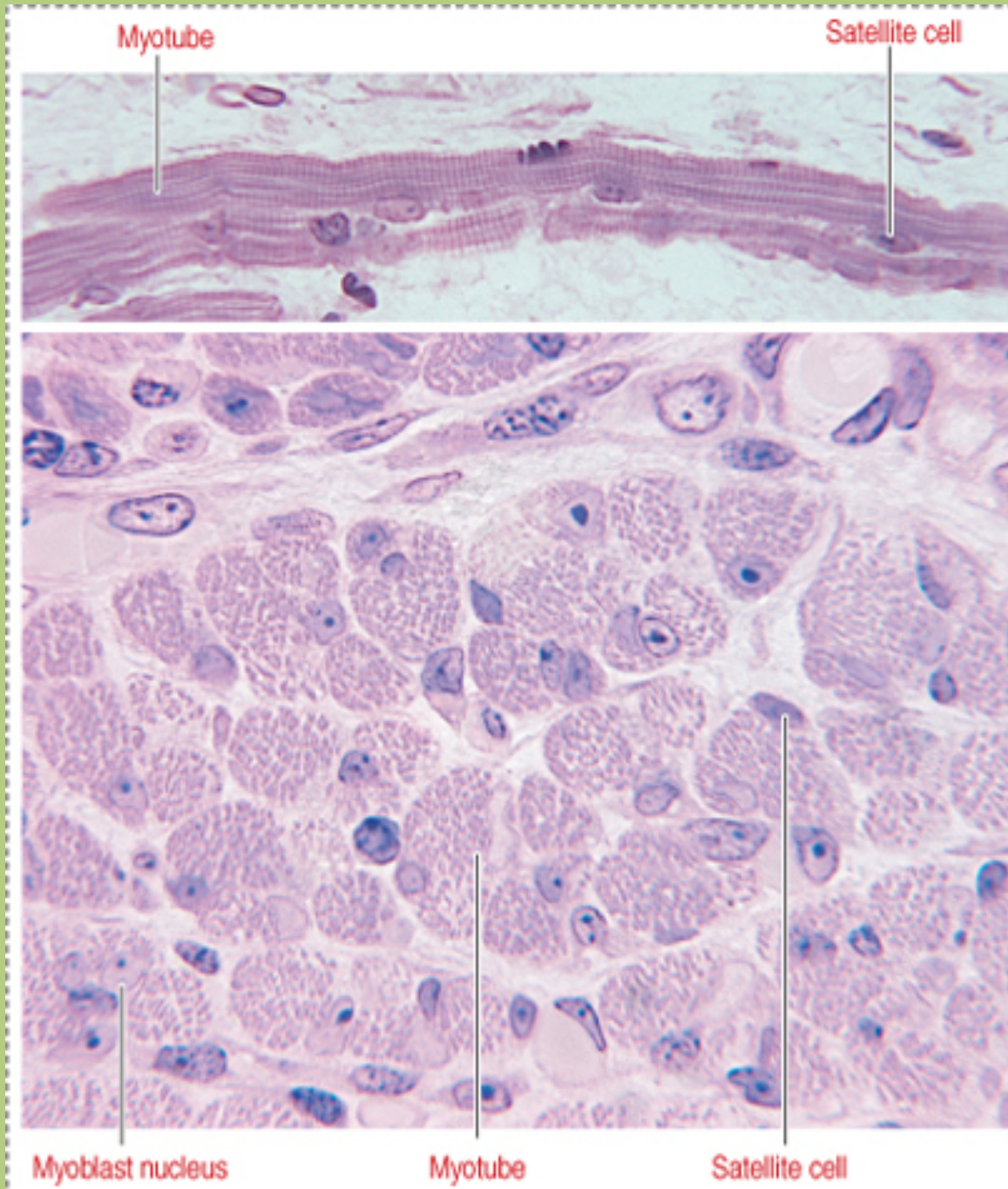
## During muscle contraction

**1** The length of the thick and thin filaments does not change. This is demonstrated by **the constant length of the A band and the distance between the Z band and the adjacent edge of the H band.**

**2** The length of the sarcomere decreases because thick and thin filaments slide past each other. This is demonstrated by **a reduction in the length of the H band and the I band.**

# REGENERATION OF STRIATED MUSCLE

- Resident muscle stem cells (=satellite cells), which upon injury activate, proliferate and fuse to repair damaged fibers.

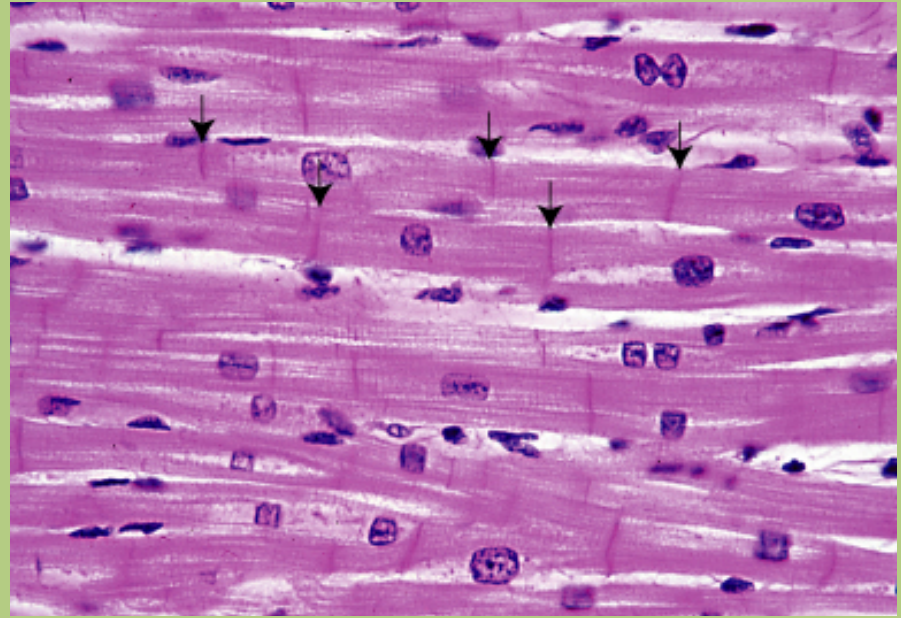


# CARDIAC MUSCLE CELLS

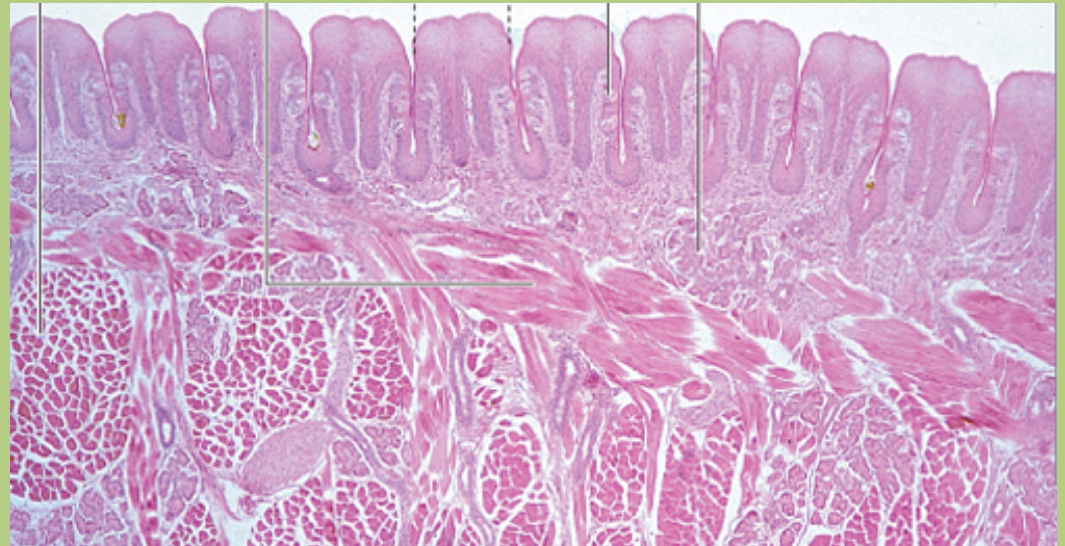
- One nucleus at the center
- Lot of mitochondria
- Actin and myosin filaments
- Well developed cell adhesion complexes (*discus intercalaris* = *Eberth's line*: *fasciae adherentes*; *macula adherens*; *gap junctions*): fast conduction
- Long-lived cells
- No regeneration

# TYPES OF STRIATED MUSCLE II.

- *Cardiac muscle*  
(arrows: Eberth's lines)



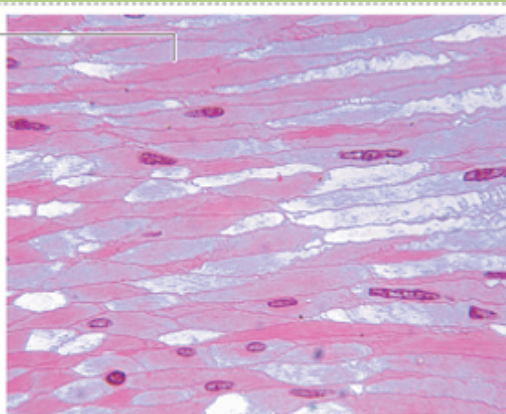
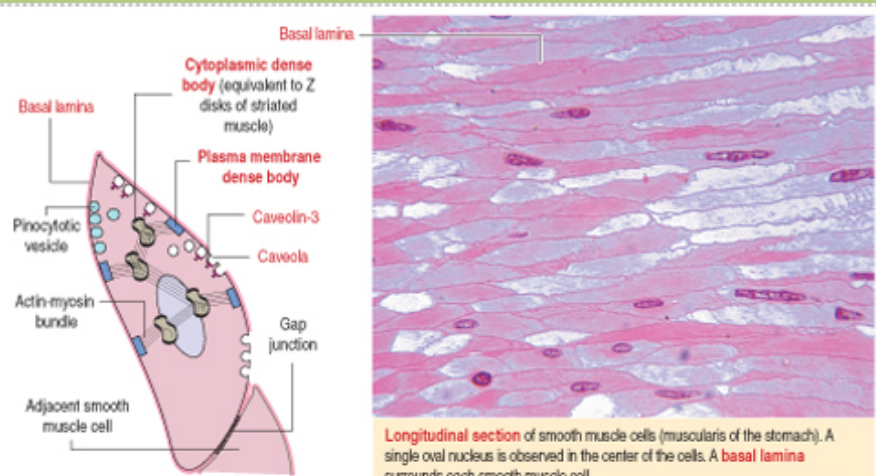
- *Visceral striated muscle*  
(tongue)



# SMOOTH MUSCLE

- Spindle-shaped cells
- Length from 20 to 200  $\mu\text{m}$
- One nucleus at the center
- The smooth muscle cells are able to synthesize elastin, collagen and other extracellular matrix's components
- No striation
- Regenerable

# SMOOTH MUSCLE



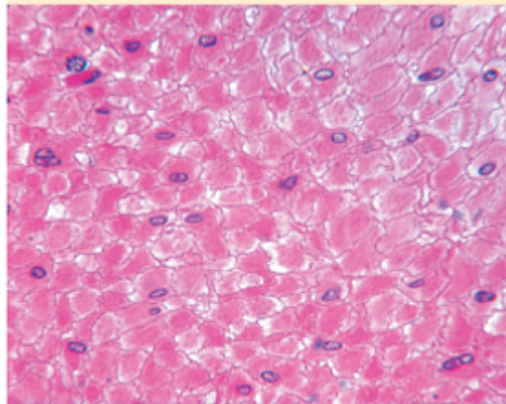
Longitudinal section of smooth muscle cells (muscularis of the stomach). A single oval nucleus is observed in the center of the cells. A basal lamina surrounds each smooth muscle cell.

## Characteristics of smooth muscle

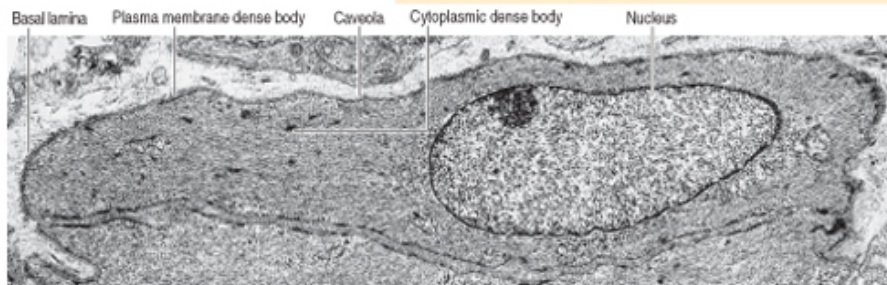
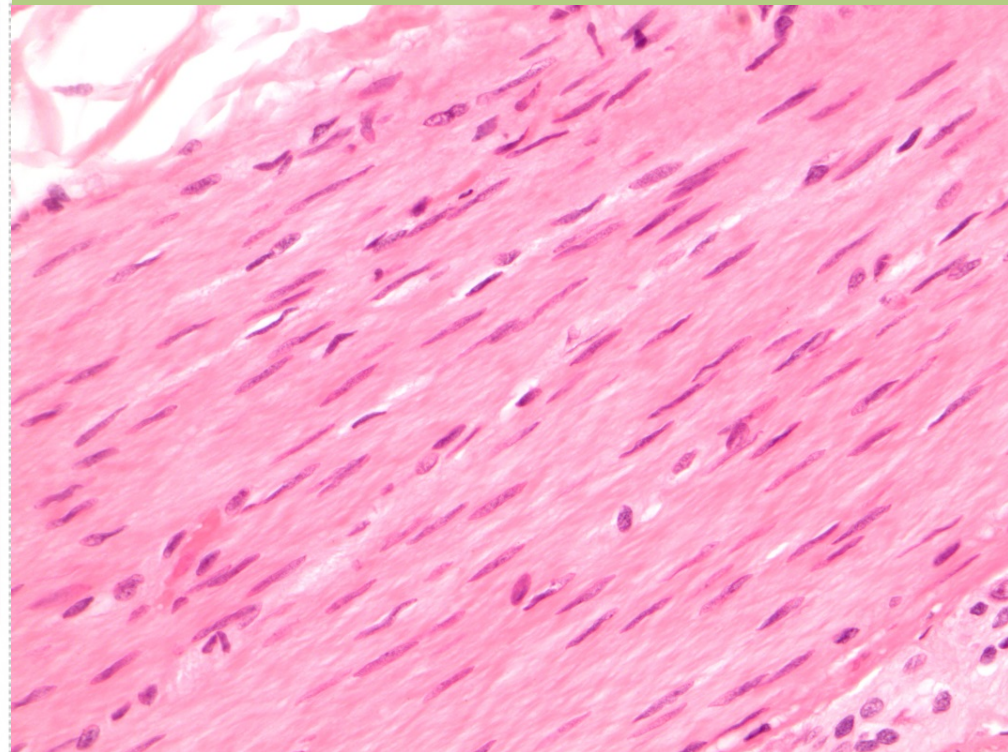
Smooth muscle is found in the walls of tubular organs, the walls of most **blood vessels**, the **iris** and **ciliary body** (eye), and **arrector pili muscle** (hair follicles), among other sites. It consists of fusiform individual cells or fibers with a **central nucleus**. Smooth cells in the walls of large blood vessels produce **elastin**.

**Caveolae**—depressions of the plasma membrane—are permanent structures involved in fluid and electrolyte transport (**pinocytosis**).

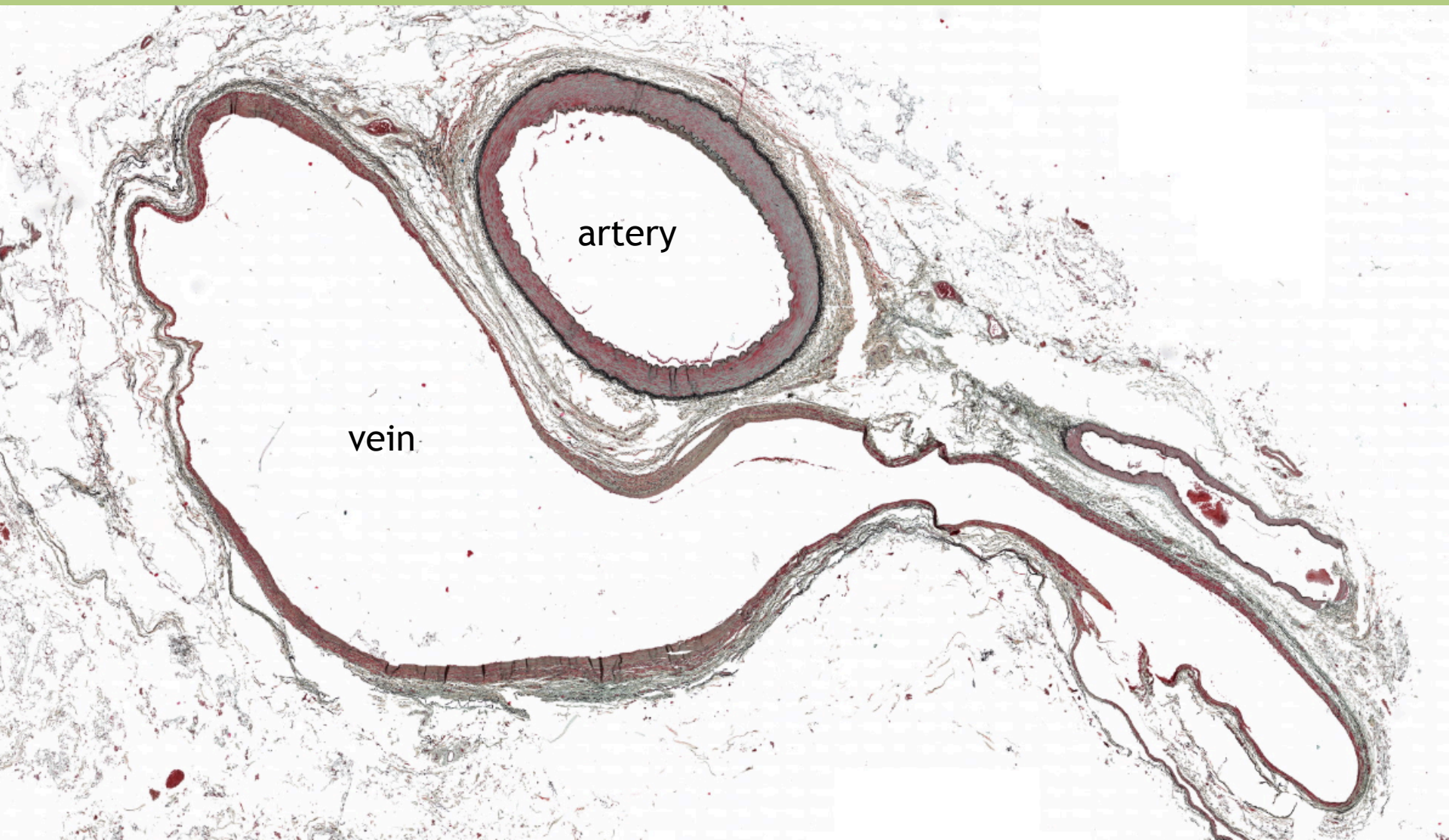
**Caveolin-3**, a protein encoded by a member of the caveolin gene family, is associated with **lipid rafts**. Complexes formed by caveolin-3 bound to **cholesterol** in a lipid raft invaginate and form caveolae. Caveolae detach from the plasma membrane to form **pinocytotic vesicles**.



Cross section of smooth muscle cells. Depending on the section level, a central nucleus is observed in some of the muscle cells.



# Blood vessels



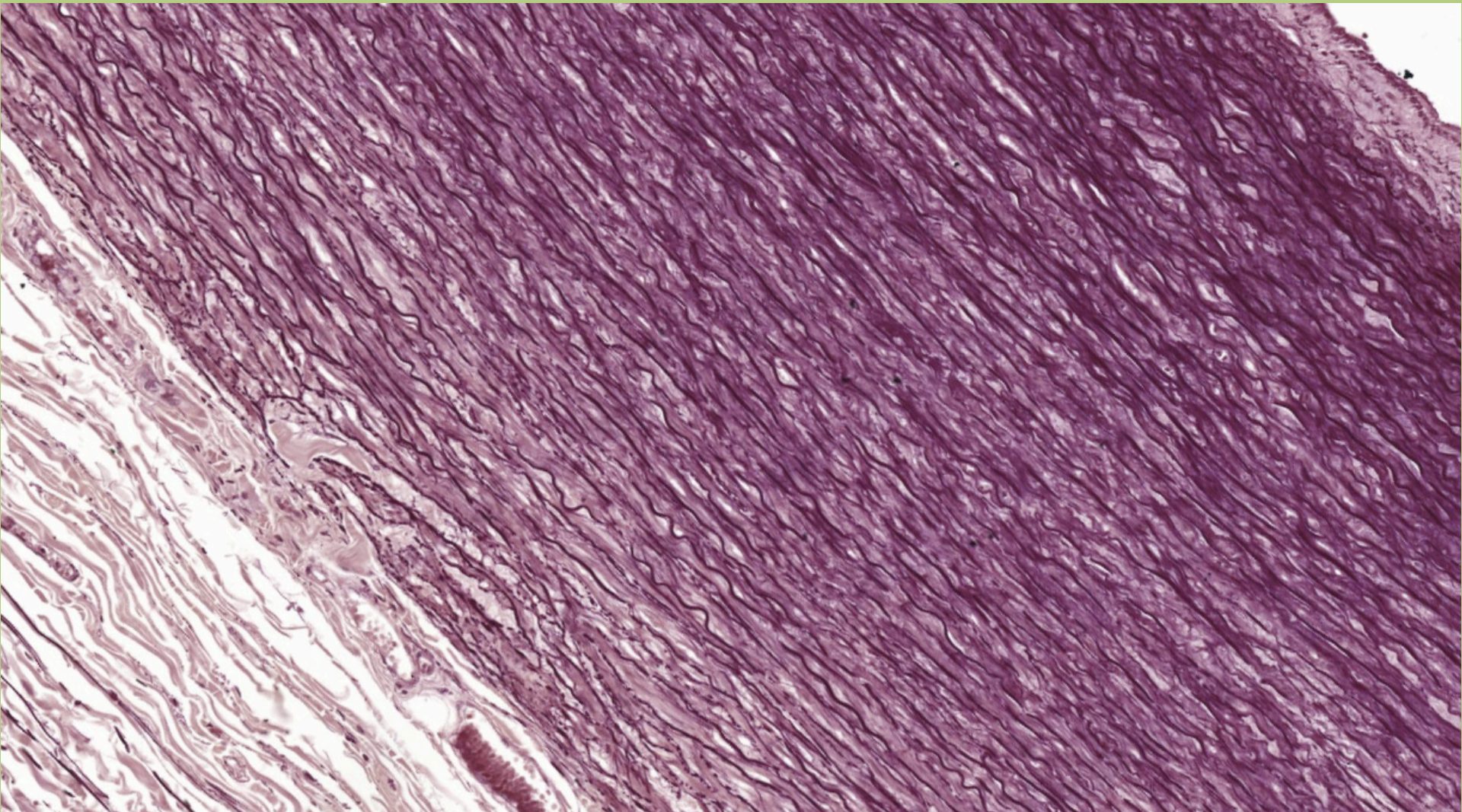
# Layers of vascular wall

**Tunica intima:** *endothelial cells*

**Tunica media:** *circumferentially arranged layers of smooth muscles cells.* In arteries, this layer is relatively thick.

**Tunica adventitia:** *outermost connective tissue layer.* In veins, it is quite thick.

# Types of vessels: elastic artery

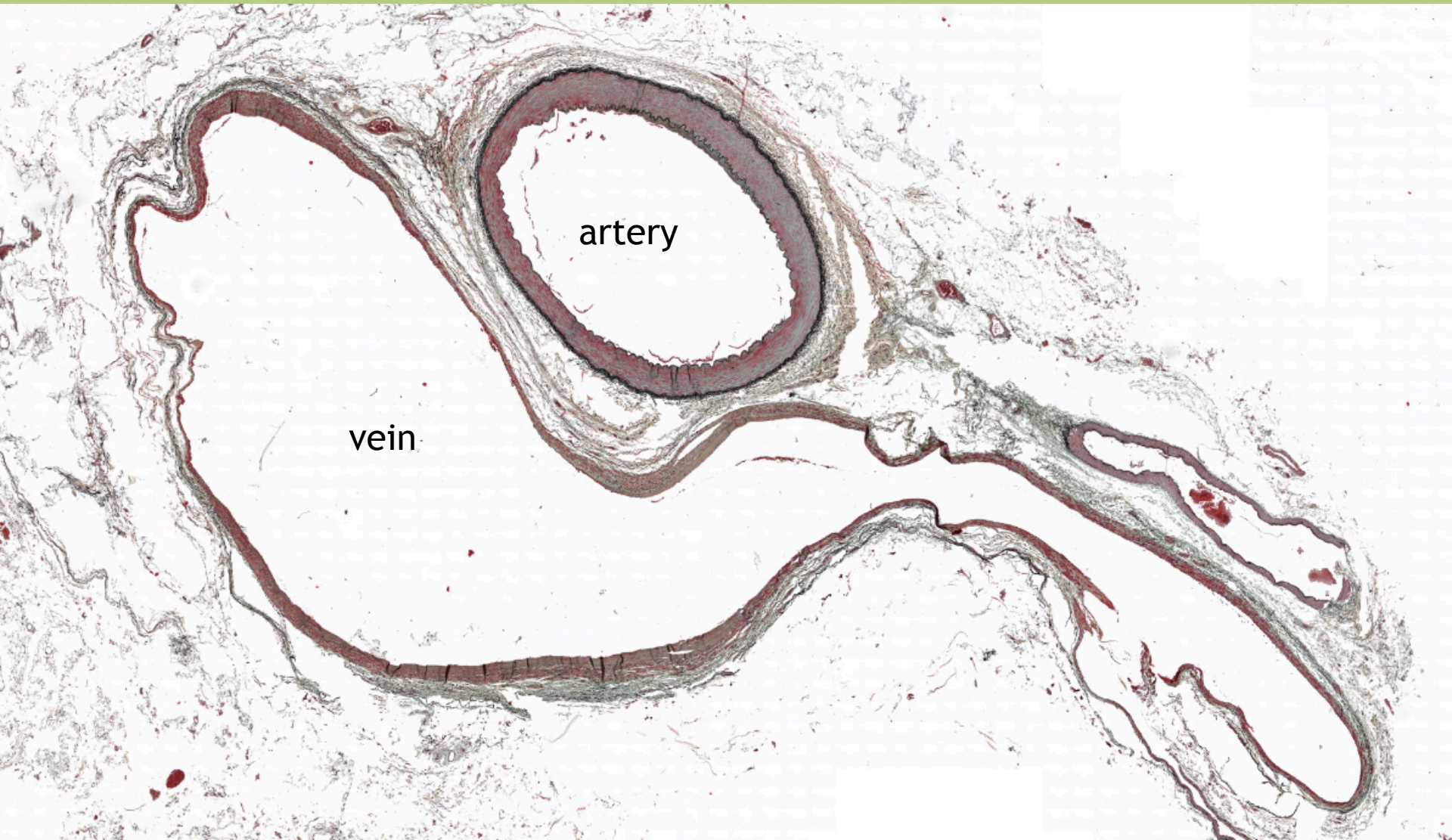


**Aorta:** the main and largest artery of the human body. Its tunica media is built up by elastic lamellae and smooth muscles cells.

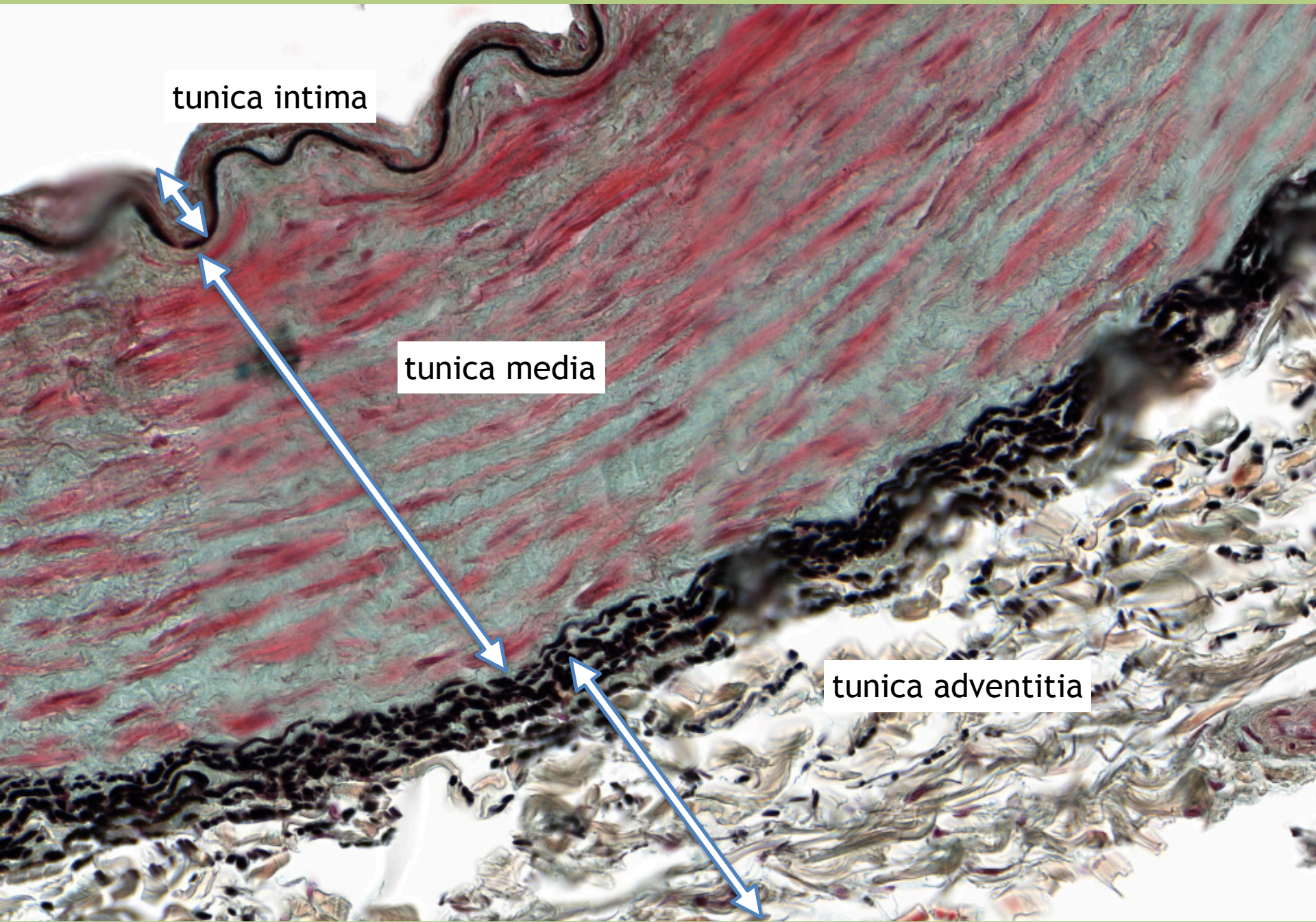
Internal elastic membrane

Vasa vasorum

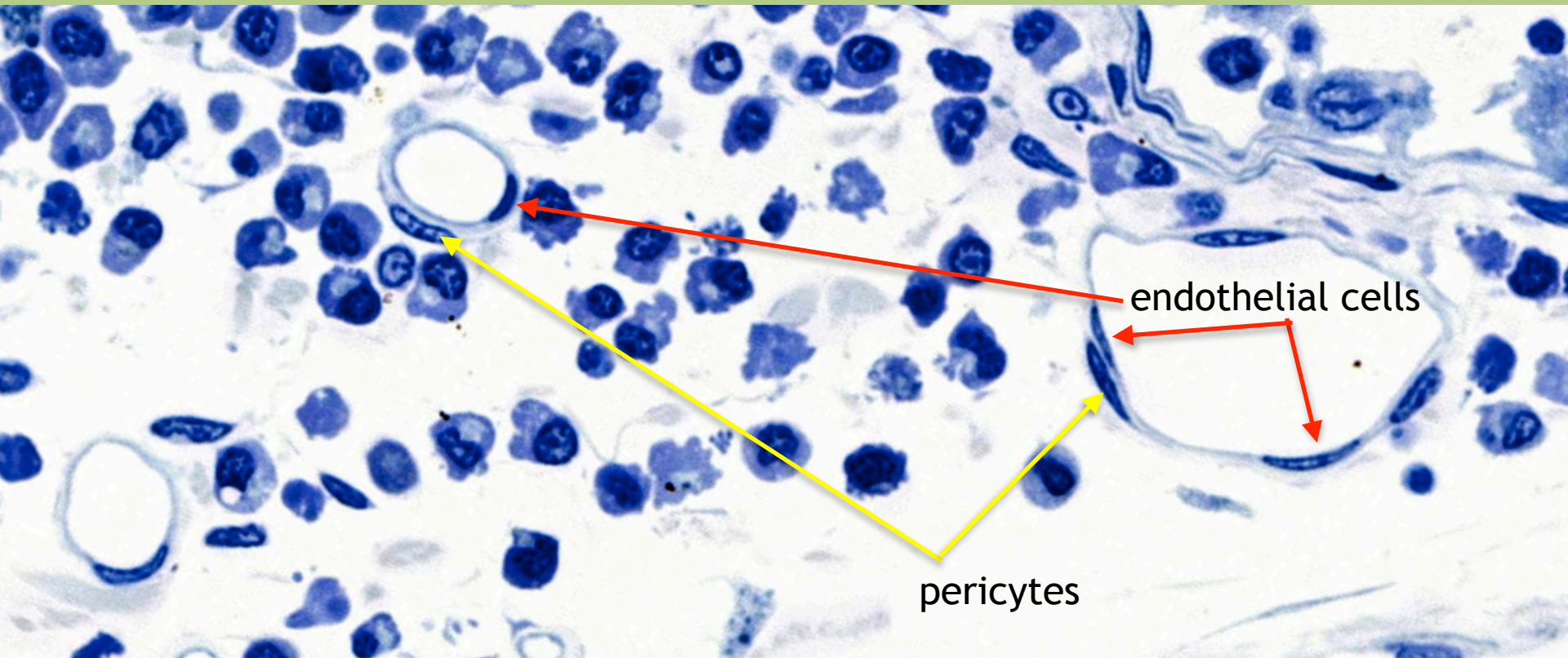
# Muscular artery



# Muscular artery



# Types of vessels: capillaries



**Pericytes:** they are contractile cells and provide support for capillaries and postcapillary venules. Histologically, pericytes display features of undifferentiated mesenchymal stem cells.

Thank you for your  
attention.

References:

McGraw-Hill Company's pictures

Gray's Anatomy for Students

Pearson Education